

# Armaghan Sarvar

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| CONTACT                 | BC Cancer Research Centre<br>Vancouver, Canada                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | [mail] <a href="mailto:asarvar@bcgsc.ca">asarvar@bcgsc.ca</a><br>[tel] (+1) 236 869 2024 | [github] <a href="https://github.com/ArmaghanSarvar">ArmaghanSarvar</a><br>[linkedin] <a href="https://www.linkedin.com/in/Armaghan-Sarvar">Armaghan-Sarvar</a><br>[web] <a href="https://armaghansarvar.github.io">armaghansarvar.github.io</a> |
| SUMMARY                 | Bioinformatics scientist specializing in machine learning, computational genomics, and epigenomics. Experienced in building end-to-end deep learning pipelines for ctDNA deconvolution, methylation sequencing, chromatin profiling, and single-cell copy number analysis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                          |                                                                                                                                                                                                                                                  |
| PROFESSIONAL EXPERIENCE | <div><b>Bioinformatics Scientist</b> December 2023–Present<br/><i>Aparicio Lab, BC Cancer Research Centre, Vancouver</i><ul style="list-style-type: none"><li>• Built end-to-end deep learning pipelines for <b>CUT&amp;Tag</b> signal modeling using Enformer and Borzoi architectures for large-scale genomic and epigenomic analyses.</li><li>• Developed a <b>ctDNA deconvolution</b> framework from plasma methylation data to infer tissue-of-origin and tumor fraction for minimal residual disease detection.</li><li>• Designed <b>5mC/hmC-aware feature extraction</b> for six-letter methylation sequencing in 184-hTERT and related cell lines, enabling epigenomic characterization at single-base resolution.</li><li>• Developed predictive models for <b>G-quadruplex</b> regulatory structures and analyzed <b>DLP+</b> single-cell data to identify copy number-based predictors of CX5461 drug response.</li><li>• Performed <b>HDP-based mutational signature decomposition</b> and SBS signature analysis of nanopore sequencing data at single-nucleotide resolution.</li></ul></div> <div><b>Machine Learning Engineer and Consultant</b> March–December 2023<br/><i>Trauma Surgery Team, Vancouver General Hospital, Vancouver</i><ul style="list-style-type: none"><li>• Improved predictive performance for clinical classification and regression problems using ensemble ML methods with hyperparameter optimization and cross-validation.</li><li>• Performed preprocessing (imputation, one-hot encoding, RFE feature selection), class imbalance handling, and outlier detection in close collaboration with the medical team.</li></ul></div> <div><b>Graduate Research Assistant</b> September 2021–December 2023<br/><i>Bioinformatics Technology Lab, Michael Smith Genome Sciences Centre, BC Cancer Agency</i><br/><i>Supervisor: Dr. Inanc Birol</i><ul style="list-style-type: none"><li>• Designed novel algorithms and data structures (Stash: stochastic tile hashing) for genome mis-assembly detection; presented at ISMB (2022) and RECOMB (2023).</li></ul></div> <div><b>Bioinformatics Research Intern</b> May–July 2022<br/><i>Silent Genomes Project, BC Children’s Hospital Research Institute, Vancouver</i><ul style="list-style-type: none"><li>• Built long-read pipelines for <b>PacBio HiFi</b> and <b>Oxford Nanopore</b> data (alignment, SNV/Indel, SV, methylation calling, STR genotyping) for intracranial aneurysm linkage analysis in an Indigenous family.</li></ul></div> <div><b>Undergraduate Research Assistant</b> March 2020–February 2021<br/><i>Amirkabir University of Technology</i> <i>Supervisor: Dr. Maryam Amirmazlaghani</i><ul style="list-style-type: none"><li>• Developed a wavelet domain defense against adversarial attacks on deep learning image classifiers, published in <i>Applied Intelligence</i> (2022).</li></ul></div> |                                                                                          |                                                                                                                                                                                                                                                  |
| EDUCATION               | <div><b>University of British Columbia</b> 2021–2023<br/>M.Sc. Bioinformatics GPA: 95.3/100 <i>Thesis: Genome Misassembly Detection using Data Structure and Algorithm Design</i></div> <div><b>Amirkabir University of Technology</b> 2016–2020<br/>B.Sc. Computer Engineering GPA: 3.75/4 <i>Thesis: Defending Deep Learning Models against</i></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                          |                                                                                                                                                                                                                                                  |

## Adversarial Attacks

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| PUBLICATIONS AND PRESENTATIONS | <ul style="list-style-type: none"> <li>● <b>Misassembly detection using Stash: A data structure based on stochastic tile hashing</b><br/>Armaghan Sarvar, Lauren Coombe, René Warren, Inanc Birol <i>RECOMB, April 2023</i></li> <li>● <b>Stash: A data structure based on stochastic tile hashing</b><br/>Armaghan Sarvar, Lauren Coombe, René Warren, Inanc Birol <i>ISMB, May 2022</i></li> <li>● <b>Defense against Adversarial Examples based on Wavelet Domain Analysis</b><br/>Armaghan Sarvar, Maryam Amirmazlaghani <i>Applied Intelligence Journal, April 2022</i></li> </ul>                                                                                                                                                                                                                                                          |  |  |
| SKILLS                         | <b>Languages:</b> Python, R, Bash, Java, C/C++, Matlab<br><b>ML &amp; DL:</b> PyTorch, TensorFlow, Scikit-learn, Hugging Face, NumPy, Pandas<br><b>Bioinformatics:</b> Snakemake, GATK, samtools, bcftools, minimap2, BWA, STAR, DESeq2, Biopython, gnomAD, HMMCopy, Mosdepth, bedtools, IGV<br><b>Sequencing:</b> Oxford Nanopore, PacBio HiFi, DLP+, CUT&Tag, Biomodal<br><b>Data &amp; Viz:</b> Matplotlib, Seaborn, Power BI, Shiny<br><b>Systems:</b> Git, Linux (Ubuntu/CentOS), HPC, MySQL, PostgreSQL                                                                                                                                                                                                                                                                                                                                    |  |  |
| SELECTED PROJECTS              | <ul style="list-style-type: none"> <li>● <b>CUT&amp;Tag Signal Modeling with Enformer and Borzoi:</b> Applied transformer-based sequence-to-function models for regulatory genomics and chromatin accessibility prediction.</li> <li>● <b>Image-to-Image Translation via Conditional GAN:</b> Built a cGAN pipeline to colorize black-and-white photographs using paired image training.</li> <li>● <b>Deep Convolutional Autoencoder:</b> Trained on the CelebA dataset with latent space visualization using T-SNE.</li> <li>● <b>U-Net for Biomedical Image Segmentation:</b> Implemented the U-Net architecture for pixel-wise segmentation of membrane image data.</li> <li>● <b>RBF Neural Network with Evolution Strategy:</b> Trained RBF networks using the DEAP library via evolutionary optimization of the cost function.</li> </ul> |  |  |
| TEACHING AND SERVICE           | <b>Mentor,</b> UBC National Canadian Medical Datathon 2025, 2026<br><b>Judge,</b> BC Cancer Research Day 2024, 2025<br><b>Judge,</b> UBC BiGSAT (Bioinformatics and Genome Science & Technology) Conference 2024, 2025<br><b>Teaching Assistant,</b> Master of Data Science, UBC 2022–2023<br><i>Communication in Data Science; Data Visualization II; Computing Platforms for Data Science</i><br><b>Organizing Committee,</b> Stemcelltalks Event 2022<br><b>Teaching Assistant,</b> Amirkabir University of Technology 2019–2020<br><i>Data Mining; Fundamentals of Image Processing; Linear Algebra</i>                                                                                                                                                                                                                                      |  |  |
| PROFESSIONAL DEVELOPMENT       | <ul style="list-style-type: none"> <li>● Project Management Fundamentals, UBC Sauder School of Business</li> <li>● Generative AI with Large Language Models, DeepLearning.AI / AWS</li> <li>● AI for Medicine Specialization, Stanford University</li> <li>● Applied Machine Learning in Python, University of Michigan</li> <li>● Neural Networks and Deep Learning, Stanford University</li> <li>● Conference Attendance: MICCAI (2023), NeurIPS (2024), ICML (2025)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                |  |  |
| LANGUAGES                      | <b>Persian:</b> Native <b>English:</b> Fluent <b>Turkish:</b> Fluent <b>Arabic:</b> Familiar                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |  |